A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Atmospheric pollution: smoke, acid rain, NOx, hydrocarbons, CO, smogs - Set 1

Mark scheme

AQA spec 3.4.3.2.1-3.4.3.2.5; Pollution; textbook pp. 245-254

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) **1** Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (930 - 250) / 250 x 100 **1** = 272% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Anticyclonic conditions:** clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled **1** Low wind velocity so air layers did not mix – London lies in a valley / basin where cold dense air collected: a temperature inversion formed **1** Smoke from coal fires was trapped below the inversion – moist air cooled to its dew point forming fog – smoke + fog = smog – the smog's high albedo prolonged the inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **temperature inversion** → a layer of warm air above cooler air that traps pollutants • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections **1** These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust	2
01.5	1 mark for the point below. **1** Clean Air Act (1956) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent gases pass through a chamber containing electrically charged wires / plates **1** Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick **1** Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **electrostatic precipitator** → charged plates that attract and collect particulates	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Effluent gases are forced to rotate / spin in a cylindrical chamber **1** The (denser) particles are thrown outwards to the chamber walls where they fall and collect – cleaned gases leave through a central pipe **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark for the point below. **1** The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Turbochargers deliver more air / oxygen to the combustion chamber **1** So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled **1** They penetrate deeper into the lungs / alveoli and are not cleared – may carry toxic / carcinogenic chemicals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	

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Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** pH 5.6 **1** Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) **1** Granite does not neutralise acids so lake water in A is acidic (4.6) **1** Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) **1** Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food – lichens and other food organisms lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1 Corrodes metals:** railway lines, railings, pipes, pylons, overhead power lines **1** Dissolves / weakens limestone and marble buildings and statues (eg York Minster) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide **1** Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO2 alone gives the weaker sulfurous acid) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Most sulfur in coal is present as (solid) iron pyrites (FeS2) **1** Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Flue gases are passed through a bed of crushed calcium carbonate (limestone) **1** Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) **1** If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide **1** The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	2
04.4	**1 mark each** 1 mark for each of the 3 points below. **1 Fluidised bed:** increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NOx (production prevented) **1 Catalytic converter:** platinum / palladium catalyst chemically reduces NOx back to nitrogen and oxygen gases in the exhaust **1** Urea sprayed into the hot flue / exhaust gases reacts with NOx, converting it to nitrogen, carbon dioxide and water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening **1** Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight **1** NO2 (from morning traffic) is broken down by UV to NO and monatomic oxygen – monatomic oxygen reacts with O2 to form ozone **1** Ozone lags behind the NO / NO2 peaks (which are produced by the morning rush-hour traffic) because the reactions take time – falls at night when there is no sunlight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure **1** NO is oxidised in the atmosphere (by oxygen / ozone) to NO2, a secondary pollutant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** PANs (peroxyacetyl nitrates) **1** Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NOx) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Heavy vehicle use giving high NOx and hydrocarbon emissions **1** Sunny climate providing the UV for photochemical reactions **1** Valley / basin topography and frequent temperature inversions trapping pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	

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Question	Marking guidance	Marks
06.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Unburnt fuel in vehicle exhaust gases **1** Evaporation from fuel tanks / spillage at filling stations **1** Solvents (paints, cleaning products) **1** Aerosol propellants **1** Gas leaks / fossil fuel exploitation (methane) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank **1** Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe **1** Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) **1** NOx are chemically reduced to nitrogen and oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues **1** Brain damage / death if enough haemoglobin is inactivated – low doses worsen chronic heart disease **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	

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Question	Marking guidance	Marks
07.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Smoke:** solid particulates (PM10, PM5, PM1) from incomplete combustion – small particles stay suspended, penetrate lungs – toxic chemicals (acids, phenols, metals) carried on particles – high albedo cools climate – blocks light reducing photosynthesis – damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths) – synergism with SO2 • **Controls:** Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion – smoke smogs have largely been eliminated in UK cities • **Acid rain gases:** SO2 (coal combustion, smelting sulfide ores) oxidised by ozone to SO3 - sulfuric acid – NOx (high temperature combustion) - nitric acid – HCl from PVC incineration – reactive gases travel long distances (transboundary pollution) • **Effects:** pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata) – dissolves exoskeletons – leaches Ca and Mg then mobilises toxic Al and Pb – corrodes metals, dissolves limestone – severity depends on bedrock (limestone neutralises) • **Controls:** fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays – switching from coal to gas / renewables	9

Do not write outside the box

Question	Marking guidance	Marks
	• **Evaluation:** SO2 emissions in Europe have fallen dramatically and lake pH is recovering – NOx harder to control because of growing vehicle numbers – catalytic converters need lead-free fuel and warm engines – FGD is costly and produces wastes / gypsum – controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO2 vs NOx **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **particulates** → fine solid particles / smoke and dust • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Smoke is particulate matter from incomplete combustion of coal, diesel, wood and wastes. Its effects follow from particle size: PM10 and smaller particles stay suspended, travel far and penetrate the lungs, carrying acids, phenols and metals; smoke also reflects sunlight (cooling), blocks light for photosynthesis and blackens buildings. Under a temperature inversion in a valley it accumulates as smog, and it acts synergistically with sulfur dioxide - the London smog of 1952 killed about 12 000 people. Acid rain gases are different: sulfur dioxide from coal combustion and sulfide smelting, and nitrogen oxides from high-temperature combustion, are reactive and soluble, forming sulfuric and nitric acids that travel long distances across borders. Low pH denatures proteins in exposed cells such as gills, eggs and root hairs, dissolves exoskeletons and limestone, corrodes metals and leaches calcium and magnesium from soils while mobilising toxic aluminium, with severity depending on bedrock.	
	Smoke control has been the clearest success: the Clean Air Act's smokeless zones, the switch to gas and electricity, electrostatic precipitators, cyclones, bag filters and diesel particulate filters have virtually eliminated smoke smogs in developed countries. Sulfur dioxide has also fallen dramatically through fuel desulfurisation, flue-gas desulfurisation and the move from coal to gas and renewables, and lake pH is recovering, although FGD is costly and produces waste. Nitrogen oxides remain the hardest: catalytic converters and fluidised-bed combustion work, but converters need unleaded fuel and warm engines and vehicle numbers keep rising. Controls have therefore succeeded for particulates and sulfur, only partly for nitrogen.	
	Weaker answer - Level 2 (5 marks)	

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Question	Marking guidance	Marks
	Smoke is made of small particles from burning coal and diesel. It causes breathing problems and smogs like in London in 1952. Acid rain is caused by sulfur dioxide and nitrogen oxides which dissolve in rain and make it acidic. Acid rain kills fish and trees and damages buildings. Smoke is controlled by smokeless zones and filters. Acid rain is controlled by taking the sulfur out of fuel and catalytic converters in cars. These methods have reduced pollution a lot, especially smoke.	
	What makes the difference	
	• Level 3 links each effect to a property (particle size, solubility, reactivity, pH denaturing proteins); Level 2 describes effects without the mechanism.	
	• Level 3 evaluates each control separately with evidence (recovering lake pH, cost of FGD, why NOx is harder); Level 2 gives one general sentence.	
	• Both answers use the 1952 smog, but Level 3 explains the conditions (inversion, synergism with SO_2) that caused it.	
07.2	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Reduces growth rates of trees / crops **1** Damages leaf cuticles (allowing SO2 to cause more damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (6400 - 220) / 6400 x 100 **1** = 96.6% (accept 97%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO2 source **1** Flue gas desulfurisation fitted to remaining coal plants – coal / oil desulfurisation **1** NOx comes mainly from road transport and vehicle numbers / distance travelled increased greatly **1** NOx is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur **1** Catalytic converters only work when warm / diesel engines produce more NOx / older vehicles remained in use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO2 (broken down by UV to monatomic oxygen which forms ozone) **1** If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 3 points below. **1** Lichens are very sensitive to acidic conditions / SO2 and different species have different tolerances **1** The species present, their abundance, size / health can be used as a biotic index of the pollution level **1** They give a long-term integrated measure of pollution rather than a single instantaneous reading – cheap, no equipment needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	

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